

软件质量保证与管理（英文）

课程名称/Course Title: 软件质量保证与管理（英文）/ Software Quality Assurance and Management

课程代码/Course Code:

任课教师/Instructor(s): 法赫德萨巴赫/ Fahad Sabah

开课学部（院）/Faculty: 计算机学院

1. 课程概要/Course Summary				
课程名称（中文） Course Title (Chinese)	软件质量保证与管理（英文）			
课程名称（英文） Course Title (English)	Software Quality Assurance and Management			
适用层次(可多选) Applicable level(s)	☒ 学硕 (Academic master)	☐ 学博 (Academic Ph.D)	☐ 专硕 (Professional master)	☐ 专博 (Professional Ph.D)
授课语言 Teaching Language	英语 English	适用学科/专业 学位类别(领域) Discipline	软件工程 Software Engineering	
学分数 Course Credit(s)	32	开课学期 Semester		
总学时 Teaching Hours in Total	共 32 学时 32 teaching hours	实验/实践学时 Hours for Experiments /Practice	共 0 学时 0 teaching hours	
预修课程要求 Pre-requisite Course(s)				
课程简介 Course Introduction (150-300 字)	这门软件质量保证与管理研究生课程旨在培养专业人士在软件生命周期的每个阶段—从需求到交付再到演进—在复杂且真实的环境中，推动质量的提升。课程融合理论与实践实践，强调现代质量工程实践、验证与确认技术、测试自动化及流程改进，均基于敏捷和 DevOps 背景。学生将设计质量策略，构建稳健的测试套件，将质量门集成到 CI/CD 流水线中，并通过迭代冲刺模拟领导端到端缺陷管理。真实案例研究和协作顶点项目反映了行业需求，同时贯穿了关于伦理质量责任（如算法公平、数据完整性和可持续测试实践）的批判性讨论。通过将高质量工程严谨性与管理敏锐度相结合，本课程将软件专业人士转变为以质量为导向的领导者，营造卓越文化，做出数据驱动的风险决策，确保在快速发展的技术生态系统中交付可靠且可扩展的解决方案。 This graduate-level course in Software Quality Assurance and Management prepares professionals to champion quality at every stage of the software lifecycle—from requirements through delivery and evolution—in complex, real-world environments. Blending theory with hands-on practice, the curriculum emphasizes modern quality engineering practices, verification and validation techniques, test			

	automation, and process improvement, all grounded in Agile and DevOps contexts. Students will design quality strategies, build robust test suites, integrate quality gates into CI/CD pipelines, and lead end-to-end defect management through iterative sprint simulations. Real-world case studies and collaborative capstone projects mirror industry demands, while critical discussions on ethical quality responsibilities—such as algorithmic fairness, data integrity, and sustainable testing practices—are woven throughout. By merging quality engineering rigor with managerial acumen, this course transforms software professionals into quality-driven leaders who foster a culture of excellence, make data-informed risk decisions, and ensure the delivery of dependable, scalable solutions in fast-moving technological ecosystems.
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2. 授课团队/Teaching group

姓名 Name	法赫德萨巴赫 Fahad Sabah	职称 Title		工作证号 ID	
姓名 Name		职称 Title		工作证号 ID	
姓名 Name		职称 Title		工作证号 ID	

3. 教学目标/Course Objective (100-200 字)

本课程通过在 SDLC 模型（瀑布、敏捷、DevOps）中嵌入保证实践，并设计从需求分析到验证、确认和发布的整体质量策略，赋能学生战略性地管理软件质量。参与者将学习制定全面的测试计划，应用测试工作量和覆盖范围的估算技术，并领导跨职能的质量举措，促进持续改进的文化。他们将掌握基于风险的测试、缺陷管理以及自动化测试框架的实施，并利用质量指标（缺陷密度、测试覆盖率、逃逸缺陷）来推动基于数据的决策。课程还涵盖伦理质量责任—包括算法公平性、数据完整性和可持续测试实践—并将工程活动与全球质量标准对齐。通过实践模拟、工具驱动的 DevOps 工作流程（将质量门集成到 CI/CD 流水线）以及毕业设计项目，毕业生将融合技术质量工程技能与管理敏锐度，在合规、AI 辅助测试等新兴趋势以及动态技术生态系统中的跨职能协作中，提供可靠且可扩展的解决方案。

This course empowers students to strategically manage software quality by embedding assurance practices across SDLC models (Waterfall, Agile, DevOps) and designing holistic quality strategies—from requirements analysis through verification, validation, and release. Participants will learn to develop comprehensive test plans, apply estimation techniques for testing effort and coverage, and lead cross-functional quality initiatives that foster a culture of continuous improvement. They will master risk-based testing, defect management, and the implementation of automated testing frameworks, as well as leverage quality metrics (defect density, test coverage, escaped defects) to drive data-informed decisions. The curriculum also addresses ethical quality responsibilities—including algorithmic fairness, data integrity, and sustainable testing practices—and aligns engineering activities with global quality standards. Through hands-on simulations, tool-driven DevOps workflows (integrating quality gates into CI/CD pipelines), and a capstone project, graduates will synthesize technical quality engineering skills with managerial acumen to deliver dependable, scalable solutions while navigating compliance, emerging trends like AI-assisted testing, and cross-functional collaboration in dynamic technological ecosystems.

4. 教学内容及进度安排/Course Content & Schedule

课次 No.	教学周 Week	教学内容 Contents	作业/实验 Assignment
1	1	软件质量保证与管理导论	案例分析讨论

		- 质量属性、QA vs QC vs QM、质量成本 - 生命周期中的质量视图（传统 vs 敏捷/DevOps） Introduction to Software Quality Assurance & Management - Quality attributes, QA vs QC vs QM, cost of quality - Quality views across lifecycles (traditional vs Agile/DevOps)	比较不同开发模型下的质量实践与挑战 Case study analysis and discussion. Compare quality practices across different development models.
2	2	质量规划与基于风险的测试策略 - 质量计划编制、验收标准、测试策略制定 - 风险识别、评估与基于风险的测试规划 - 测试范围界定与工作分解（Test WBS） Quality Planning & Risk-Based Test Strategy - Quality planning, acceptance criteria, test strategy - Risk identification, assessment, and risk-based test planning - Test scope definition and Test WBS	为示例项目创建基于风险的测试计划与 Test WBS Create a risk-based test plan and Test WBS for a sample project.
3	3	验证与确认技术 - 静态测试：审查、走查、静态分析 - 动态测试技术：等价类划分、边界值分析、状态迁移测试 - 白盒测试与探索性测试简介 Verification & Validation Techniques - Static testing: reviews, walkthroughs, static analysis - Dynamic techniques: equivalence partitioning, boundary value, state transition - White-box testing and exploratory testing introduction	测试用例设计与执行 对给定模块进行黑盒与白盒测试用例设计并模拟执行 Test case design and execution. Design black-box and white-box test cases for a given module.
4	4	测试管理与敏捷测试实践 - 测试生命周期、测试进度与估算（测试点估算） - 敏捷测试四象限、Scrum/Kanban 中的测试角色 - 测试管理工具与缺陷生命周期（JIRA/Xray 示例） Test Management & Agile Testing Practices - Test lifecycle, test scheduling and estimation (test point estimation) - Agile Testing Quadrants, tester roles in Scrum/Kanban - Test management tools and defect lifecycle (JIRA/Xray examples)	使用 JIRA 模拟一个 Sprint 的测试管理与缺陷追踪 Simulate test management and defect tracking for one sprint using JIRA.
5	5	测试自动化与持续质量工程 - 测试自动化策略、框架选型（Selenium 等） - CI/CD 流水线中的质量门与持续测试 - 容器化测试环境与基础设施即代码的验证 Test Automation & Continuous Quality Engineering - Test automation strategy, framework selection (Selenium, etc.)	配置一条包含自动化测试质量门的 Jenkins 流水线 Configure a Jenkins pipeline with automated test quality gates.

		• Quality gates in CI/CD pipelines and continuous testing • Containerized test environments and Infrastructure-as-Code validation	
6	6	质量度量、分析与改进 • 核心质量度量：缺陷密度、测试覆盖率、逃逸缺陷、MTTR • 质量分析技术与根本原因分析（RCA） • 构建质量度量仪表盘（Power BI/Grafana） Quality Metrics, Analysis & Improvement • Core metrics: defect density, test coverage, escaped defects, MTTR • Quality analysis techniques and root cause analysis (RCA) • Building quality dashboards (Power BI/Grafana)	基于历史数据构建质量度量仪表盘并撰写分析报告 Build a quality dashboard from historical data and write an analysis report.
7	7	过程改进、标准与质量文化 • 过程改进模型与标准：CMMI、ISO 25010/9001、TMMi • 技术债务管理与质量技术债量化 • 质量文化、伦理责任（AI 公平性、数据隐私）与合规 Process Improvement, Standards & Quality Culture • Improvement models and standards: CMMI, ISO 25010/9001, TMMi • Technical debt management and quality debt quantification • Quality culture, ethical responsibility (AI fairness, data privacy), and compliance	评估一个真实组织中的软件质量成熟度并提出改进路线图 Assess the software quality maturity of a real organization and propose an improvement roadmap.
8	8	质量保证与管理毕业项目 • 制定完整的质量保证计划（含风险分析、测试策略、度量与改进） • 在给定系统中实施质量门与关键度量 • 成果汇报：20 分钟小组演示（含问答环节） Capstone Project in Software Quality Assurance & Management • Develop a complete quality assurance plan (risk, test strategy, metrics, improvement) • Implement quality gates and key metrics on a given system • Presentation: 20-minute group presentation with Q&A	毕业项目综合实践 提交质量保证方案并完成演示 Integrated capstone project. Submit a quality assurance plan and deliver the final presentation.
5. 课程考核及成绩评定/Course Assessment & Grading			
考核指标* Assessment Criteria	权重 Percentage	评定标准 Assessment Standard	
课堂测验 Quizzes	10%	通过阶段性选择题测验考查学生对软件质量属性、QA/QC/QM、风险测试、验证与确认技术、敏捷测试、	

		测试自动化、质量指标及过程改进模型等核心知识的掌握程度。主要评价概念理解的准确性、指标计算的正确性、实际情境分析能力及知识综合应用能力。 Periodic MCQ quizzes assess students' understanding of quality attributes, QA/QC/QM, risk-based testing, verification and validation, Agile testing, test automation, quality metrics, and process improvement models. Assessment focuses on conceptual accuracy, correctness of metric calculations, analysis of practical scenarios, and integrated application of course knowledge.
作业/实验 Assignment(s)	35	通过四项阶段性任务评价学生将课程知识应用于软件项目质量管理活动的的能力，包括质量规划与风险测试方案、验证与确认及敏捷测试实践、测试自动化与持续质量工程，以及质量度量与过程改进分析。 Four progressive tasks assess students' ability to apply course knowledge to software project quality-management activities, including quality planning and risk-based testing, verification and validation with Agile testing practices, test automation and continuous quality engineering, and quality metrics with process improvement analysis.
课程考试 Course Exam(s).	25	综合考查学生对课程理论、方法、技术和标准的掌握程度。考试内容包括基本概念辨析、质量指标计算、测试技术应用、风险分析、缺陷与技术债务分析，以及软件质量改进方案设计。主要评价知识掌握的准确性、分析推理能力和综合问题解决能力。 The examination comprehensively assesses students' mastery of course theories, methods, techniques, and standards. It includes conceptual questions, quality-metric calculations, application of test techniques, risk analysis, defect and technical-debt analysis, and software quality improvement planning. Assessment focuses on accuracy, analytical reasoning, and integrated problem-solving ability.
Project	30	以 2 - 3 人为一组，围绕选定的软件项目完成综合质量档案，并进行最终展示与答辩。评价内容包括质量目标与范围、风险测试策略、测试设计与执行、缺陷管理、自动化测试、CI/CD 质量门、质量指标仪表盘、技术债务、成熟度评估和改进路线图。成绩同时考虑项目成果质量、团队协作及个人贡献。 Working in groups of two to three, students develop a comprehensive Software Project Quality Portfolio for a selected software project and complete a final presentation and defence. Assessment covers quality

		objectives and scope, risk-based test strategy, test design and execution, defect management, test automation, CI/CD quality gates, quality dashboards, technical debt, maturity assessment, and an improvement roadmap. Grades consider project quality, teamwork, and individual contribution.				
* 各项考核指标可自由设置，总权重为 100%。						
6. 教材/Textbook(s) （如使用自编讲义，请在“名称”列中备注说明）						
序号 No.	名称 Title	作者 Author(s)	标准书号 ISBN	出版机构 Publisher	出版日期 Publication Date	是否必读 Mandatory or Elective
1	Software Engineering: A Practitioner’s Approach	Roger Pressman and Bruce Maxim	9781260548006	McGraw-Hill Education	2019	Elective
2						
7.教学参考资料/Reading Materials and References						
序号 No.	名称 Title	作者 Author(s)	标准号码 ISBN/DIO	出版机构 Publisher	出版日期 Publication Date	是否必读 Mandatory or Elective
1						